LESSON 01 / 03

The vector triple product returns to a plane

The **BAC–CAB** identity is

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b}).$$

The result is a linear combination of $\mathbf{b},\mathbf{c}$, as it must be: a vector perpendicular to $\mathbf{b}\times\mathbf{c}$ lies in their span. The identity can be proved by linearity and checking basis vectors, or by expanding components once.

The other grouping gives

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{a}(\mathbf{b}\cdot\mathbf{c}).$$

The two expressions are different in general. Cross products need parentheses.

WORKED EXAMPLE

Two groupings, two different answers

For $\mathbf{a}=(7,8,9)$, $\mathbf{b}=(1,2,3)$ and $\mathbf{c}=(4,5,6)$,

$$\mathbf{a}\cdot\mathbf{c}=122,\qquad \mathbf{a}\cdot\mathbf{b}=50,\qquad \mathbf{b}\cdot\mathbf{c}=32.$$

Hence

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=122\mathbf{b}-50\mathbf{c}=(-78,\,-6,66),$$

whereas

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=122\mathbf{b}-32\mathbf{a}=(-102,\,-12,78).$$

They are unequal, but each expansion agrees with direct cross-product computation.

COMMON TRAP

Not associative does not mean never equal

The two groupings may coincide for special dependent or orthogonal vectors. "Not associative" means no universal equality is available. Prove a special case if you want to remove parentheses; never remove them by habit.

Jacobi's identity is three expansions cancelling

For every $\mathbf{a},\mathbf{b},\mathbf{c}$,

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})+\mathbf{b}\times(\mathbf{c}\times\mathbf{a})+\mathbf{c}\times(\mathbf{a}\times\mathbf{b})=\mathbf{0}.$$

SHORT DERIVATION

Cancellation by cyclic terms

Apply BAC–CAB to each term:

$$\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b})+\mathbf{c}(\mathbf{b}\cdot\mathbf{a})-\mathbf{a}(\mathbf{b}\cdot\mathbf{c})+\mathbf{a}(\mathbf{c}\cdot\mathbf{b})-\mathbf{b}(\mathbf{c}\cdot\mathbf{a}).$$

Commutativity of the dot product makes corresponding vector coefficients cancel in pairs. The numerical example above therefore gives a zero cyclic sum even though neither individual triple product is zero.

A scalar quadruple product generalizes Lagrange

Another useful identity is

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d})=(\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c}).$$

Setting $\mathbf{c}=\mathbf{a}$ and $\mathbf{d}=\mathbf{b}$ recovers Lagrange's identity. This form is valuable when two planes or two oriented areas must be compared without computing every cross component separately.

Keep the triple-product shapes distinct

Expression	Type	Main interpretation
a·(b×c)	scalar	signed volume / dependence
a×(b×c)	vector	BAC–CAB in span(b,c)
(a×b)·(c×d)	scalar	dot of two oriented normals
cyclic Jacobi sum	zero vector	three expansions cancel

READ THE FIGURE

First identify whether the answer is a scalar or a vector. That simple type check rejects many incorrect identities immediately.

WORKED EXAMPLE

A quick scalar-quadruple check

Let $\mathbf{a}=\mathbf{i}$, $\mathbf{b}=\mathbf{j}$, $\mathbf{c}=\mathbf{i}+\mathbf{j}$ and $\mathbf{d}=\mathbf{j}+\mathbf{k}$. Then

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d})=\mathbf{k}\cdot(\mathbf{i}-\mathbf{j}+\mathbf{k})=1.$$

The right side is $(\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})=1\cdot1-0\cdot1=1$.

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Barycentric coordinates are vector weights and area ratios

For a nondegenerate triangle ABC, normalized barycentric coordinates write

$$P = \alpha A + \beta B + \gamma C, \qquad \alpha + \beta + \gamma = 1.$$

For a point inside the triangle,

$$\alpha = \frac{[PBC]}{[ABC]}, \qquad \beta = \frac{[PCA]}{[ABC]}, \qquad \gamma = \frac{[PAB]}{[ABC]}.$$

Outside the triangle, use signed areas and signed weights. Unnormalized coordinates $(\lambda_1 : \lambda_2 : \lambda_3)$ become normalized after division by their nonzero sum.

WORKED EXAMPLE

Coordinates from areas and back again

For $A = (0, 0)$, $B = (4, 0)$, $C = (0, 3)$ and $P = (1, 1)$,

$$P = \frac{5}{12}A + \frac{1}{4}B + \frac{1}{3}C.$$

Thus the unnormalized coordinates are $(5 : 3 : 4)$. Also $[ABC] = 6$ and $[PBC] = 5/2$, so $[PBC]/[ABC] = 5/12$, matching the coefficient of A. The three subtriangle area ratios are $5/12, 1/4, 1/3$, whose sum is 1.

Ceva is a product of directed ratios

Put

$$x = \frac{BX}{XC}, \qquad y = \frac{CY}{YA}, \qquad z = \frac{AZ}{ZB},$$

where $X \in BC$, $Y \in CA$ and $Z \in AB$. The cevians AX, BY, CZ are concurrent exactly when

$$xyz = 1,$$

with directed ratios used for external points. In barycentrics, $X = (0 : 1 : x)$, $Y = (y : 0 : 1)$ and $Z = (1 : z : 0)$; eliminating the cevian equations gives the same product condition.

WORKED EXAMPLE

A concurrent Ceva configuration

If $BX : XC = 2 : 1$, $CY : YA = 3 : 1$ and $AZ : ZB = 1 : 6$, then

$$x = 2, \qquad y = 3, \qquad z = \frac{1}{6}, \qquad xyz = 1.$$

Therefore AX, BY, CZ are concurrent. No coordinates of the intersection are required for the conclusion.

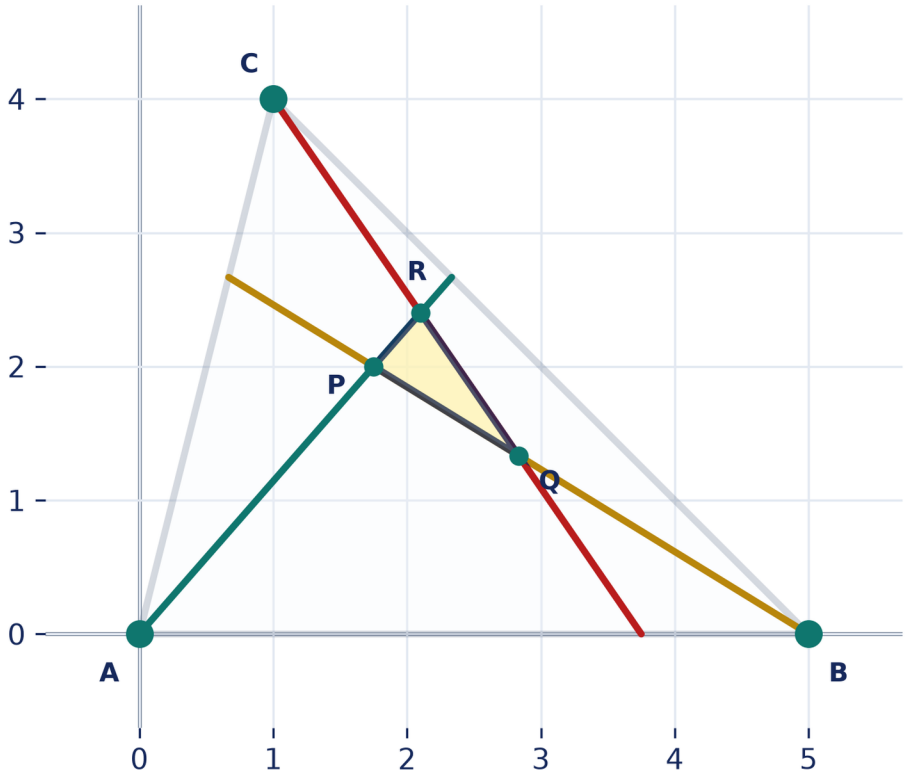
Routh measures the triangle bounded by three cevians

With the same ratio convention and $xyz \neq 1$, the cevians bound a triangle PQR satisfying

$$\frac{[PQR]}{[ABC]} = \frac{(xyz-1)^2}{(xy+x+1)(yz+y+1)(zx+z+1)}.$$

When $xyz = 1$, the cevians concur and the numerator correctly becomes zero.

The three cevians bound the central triangle PQR



READ THE FIGURE
A=(0,0), B=(5,0), C=(1,4), x=2, y=1/2, z=3. Direct cevian intersections give P=(7/4,2), Q=(17/6,4/3), R=(21/10,12/5).

WORKED EXAMPLE

An exact Routh calculation, checked by intersections

For $x = 2$, $y = 1/2$ and $z = 3$, we have $xyz = 3$ and

$$xy + x + 1 = 4, \qquad yz + y + 1 = 3, \qquad zx + z + 1 = 10.$$

Therefore

$$\frac{[PQR]}{[ABC]} = \frac{(3-1)^2}{4 \cdot 3 \cdot 10} = \frac{1}{30}.$$

The displayed coordinates independently give the same ratio. Since $[ABC] = 10$, the central triangle has area $1/3$. Keep the cyclic factors in order: here $yz + y + 1 = 3/2 + 1/2 + 1 = 3$, not 2; using 2 would give the incorrect result $1/20$.

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Choose an origin that makes the theorem disappear

Vector proofs become short when the origin matches the symmetry. For a parallelogram, use vertices $0, \mathbf{a}, \mathbf{a}+\mathbf{b}, \mathbf{b}$. For a triangle's medians, put one vertex at the origin. For a circumcentre problem, put the circumcentre at the origin.

WORKED EXAMPLE

Varignon's parallelogram in one subtraction

Let A, B, C, D be any four points in a plane, and let P, Q, R, S be the midpoints of AB, BC, CD, DA. Then

$$Q - P = \frac{B + C - A - B}{2} = \frac{C - A}{2},$$

$$R - S = \frac{C + D - D - A}{2} = \frac{C - A}{2}.$$

Thus $PQ \parallel SR$ and $PQ = SR$; similarly $QR \parallel PS$. The midpoint quadrilateral is always a parallelogram, even when the original quadrilateral is not one.

With the circumcentre at O = 0, H = A+B+C

Suppose the circumcentre O of $\triangle ABC$ is the origin, so $|A| = |B| = |C| = R$. Define $H = A + B + C$. Then

$$(H - A) \cdot (C - B) = (B + C) \cdot (C - B) = |C|^2 - |B|^2 = 0.$$

Thus $AH \perp BC$; cyclic relabelling gives the other two altitudes. Hence H is the orthocentre.

The centroid is $G = (A + B + C)/3 = H/3$. Therefore O, G, H are collinear and

$$OG : GH = 1 : 2.$$

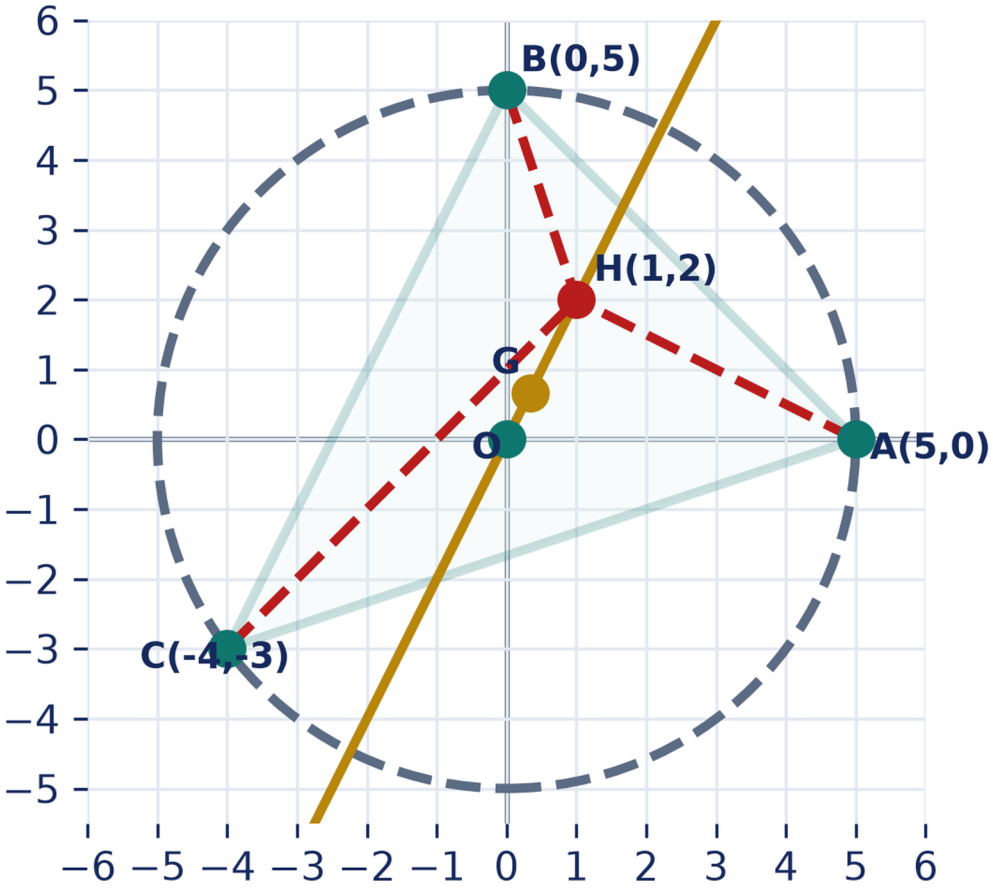
With a general circumcentre O, translate back to obtain $H = A + B + C - 2O$ and $G - O = (H - O)/3$.

COMMON TRAP

Equal distances from a sphere centre are not enough in 3D

For three points in space, a triangle's circumcentre must lie **in the triangle's plane** as well as be equidistant from its vertices. A sphere centre equidistant from all three need not lie in that plane. Before using $H = A + B + C$ with $O = 0$, verify both the equal lengths and that O belongs to the plane of A, B, C.

O, G and H lie on one Euler line



READ THE FIGURE
All three vertices lie on x^2+y^2=25. H=A+B+C=(1,2), G=H/3, and the dashed segments are altitudes.

Verify the example with lengths and dot products

For $A = (5, 0)$, $B = (0, 5)$ and $C = (-4, -3)$,

$$|A|^2 = |B|^2 = |C|^2 = 25,$$

so the origin is genuinely the circumcentre. Set $H = (1, 2)$. Then

$$(H - A) \cdot (C - B) = (-4, 2) \cdot (-4, -8) = 16 - 16 = 0,$$

$$(H - B) \cdot (A - C) = (1, -3) \cdot (9, 3) = 9 - 9 = 0.$$

The third altitude follows by the same calculation. Also $G = (1/3, 2/3) = H/3$, so

$$OG^2 = \tfrac{5}{9}, \qquad GH^2 = \tfrac{20}{9} = 4OG^2.$$

This verifies both collinearity and $OG : GH = 1 : 2$ without relying on the drawing.